

```

explain analyze
select
  (
    select product_category
    from (
      select
        p.product_category,
        count(*) as orders_count
      from opt_orders o
      join opt_clients c
        on c.id = o.client_id
      join opt_products p
        on p.product_id = o.product_id
      where c.status = 'active'
        and extract(year from o.order_date) = 2023
      group by p.product_category
    ) category_stats_1
    order by orders_count desc, product_category asc
    limit 1
  ) as most_popular_category,
  (
    select product_category
    from (
      select
        p.product_category,
        count(*) as orders_count
      from opt_orders o
      join opt_clients c
        on c.id = o.client_id
      join opt_products p
        on p.product_id = o.product_id
      where c.status = 'active'
        and extract(year from o.order_date) = 2023
      group by p.product_category
    ) category_stats_2
    order by orders_count asc, product_category asc
    limit 1
  ) as least_popular_category,
  (

```

```

select avg(orders_count)
from (
    select
        p.product_category,
        count(*) as orders_count
    from opt_orders o
    join opt_clients c
        on c.id = o.client_id
    join opt_products p
        on p.product_id = o.product_id
    where c.status = 'active'
        and extract(year from o.order_date) = 2023
    group by p.product_category
) category_stats_3
) as avg_orders_per_category

```

QUERY PLAN

Result (cost=53028.65..53028.66 rows=1 width=268) (actual time=2649.312..2674.686 rows=1.00 loops=1)

Buffers: shared hit=27957

InitPlan 1

-> Limit (cost=17676.20..17676.20 rows=1 width=18) (actual time=1518.498..1527.697 rows=1.00 loops=1)

Buffers: shared hit=9319

-> Sort (cost=17676.20..17676.21 rows=5 width=18) (actual time=1518.495..1527.692 rows=1.00 loops=1)

Sort Key: category_stats_1.orders_count DESC, category_stats_1.product_category

Sort Method: top-N heapsort Memory: 25kB

Buffers: shared hit=9319

-> Subquery Scan on category_stats_1 (cost=17664.09..17676.17 rows=5 width=18) (actual time=1508.983..1527.503 rows=5.00 loops=1)

Buffers: shared hit=9319

-> Finalize GroupAggregate (cost=17664.09..17676.17 rows=5 width=18) (actual time=1508.974..1527.487 rows=5.00 loops=1)

Group Key: p.product_category
Buffers: shared hit=9319
-> Gather Merge (cost=17664.09..17676.08 rows=8 width=18) (actual time=1505.785..1527.452 rows=10.00 loops=1)
Workers Planned: 1
Workers Launched: 1
Buffers: shared hit=9319
-> Partial GroupAggregate (cost=16664.08..16675.17 rows=5 width=18) (actual time=1390.096..1401.634 rows=5.00 loops=2)
Group Key: p.product_category
Buffers: shared hit=9319
-> Sort (cost=16664.08..16667.76 rows=1473 width=10) (actual time=1385.571..1390.574 rows=49933.50 loops=2)
Sort Key: p.product_category
Sort Method: quicksort Memory: 1537kB
Buffers: shared hit=9319
Worker 0: Sort Method: quicksort Memory: 1537kB
-> Hash Join (cost=13677.54..16586.56 rows=1473 width=10) (actual time=820.789..1332.701 rows=49933.50 loops=2)
Hash Cond: (o.product_id = p.product_id)
Buffers: shared hit=9311
-> Parallel Hash Join (cost=13629.04..16534.19 rows=1473 width=4) (actual time=812.637..1298.839 rows=49933.50 loops=2)
Hash Cond: (c.id = o.client_id)
Buffers: shared hit=9259
-> Parallel Seq Scan on opt_clients c (cost=0.00..2641.29 rows=29453 width=16) (actual time=0.353..434.733 rows=25012.00 loops=2)
Filter: ((status)::text = 'active'::text)

Rows Removed by Filter: 24988
Buffers: shared hit=1906
-> Parallel Hash (cost=13603.00..13603.00 rows=2083 width=20) (actual time=811.938..811.939 rows=99852.50 loops=2)
Buckets: 262144 (originally 8192) Batches: 1 (originally 1) Memory Usage: 15008kB
Buffers: shared hit=7353
-> Parallel Seq Scan on opt_orders o (cost=0.00..13603.00 rows=2083 width=20) (actual time=1.414..713.415 rows=99852.50 loops=2)
Filter: (EXTRACT(year FROM order_date) = '2023'::numeric)
Rows Removed by Filter: 400148
Buffers: shared hit=7353
-> Hash (cost=36.00..36.00 rows=1000 width=14) (actual time=7.765..7.766 rows=1000.00 loops=2)
Buckets: 1024 Batches: 1 Memory Usage: 55kB
Buffers: shared hit=52
-> Seq Scan on opt_products p (cost=0.00..36.00 rows=1000 width=14) (actual time=1.957..6.940 rows=1000.00 loops=2)
Buffers: shared hit=52
InitPlan 2
-> Limit (cost=17676.20..17676.20 rows=1 width=18) (actual time=538.763..546.898 rows=1.00 loops=1)
Buffers: shared hit=9319
-> Sort (cost=17676.20..17676.21 rows=5 width=18) (actual time=538.758..546.891 rows=1.00 loops=1)
Sort Key: category_stats_2.orders_count, category_stats_2.product_category
Sort Method: top-N heapsort Memory: 25kB
Buffers: shared hit=9319
-> Subquery Scan on category_stats_2 (cost=17664.09..17676.17 rows=5 width=18) (actual time=529.255..546.715 rows=5.00 loops=1)

Buffers: shared hit=9319
-> Finalize GroupAggregate (cost=17664.09..17676.17 rows=5 width=18) (actual time=529.252..546.705 rows=5.00 loops=1)
Group Key: p_1.product_category
Buffers: shared hit=9319
-> Gather Merge (cost=17664.09..17676.08 rows=8 width=18) (actual time=526.084..546.667 rows=10.00 loops=1)
Workers Planned: 1
Workers Launched: 1
Buffers: shared hit=9319
-> Partial GroupAggregate (cost=16664.08..16675.17 rows=5 width=18) (actual time=450.861..464.311 rows=5.00 loops=2)
Group Key: p_1.product_category
Buffers: shared hit=9319
-> Sort (cost=16664.08..16667.76 rows=1473 width=10) (actual time=447.336..452.919 rows=49933.50 loops=2)
Sort Key: p_1.product_category
Sort Method: quicksort Memory: 1537kB
Buffers: shared hit=9319
Worker 0: Sort Method: quicksort Memory: 1537kB
-> Hash Join (cost=13677.54..16586.56 rows=1473 width=10) (actual time=319.635..402.070 rows=49933.50 loops=2)
Hash Cond: (o_1.product_id = p_1.product_id)
Buffers: shared hit=9311
-> Parallel Hash Join (cost=13629.04..16534.19 rows=1473 width=4) (actual time=318.203..380.707 rows=49933.50 loops=2)
Hash Cond: (c_1.id = o_1.client_id)
Buffers: shared hit=9259

-> Parallel Seq Scan on opt_clients c_1 (cost=0.00..2641.29 rows=29453 width=16) (actual time=0.024..23.366 rows=25012.00 loops=2)
Filter: ((status)::text = 'active'::text)
Rows Removed by Filter: 24988
Buffers: shared hit=1906
-> Parallel Hash (cost=13603.00..13603.00 rows=2083 width=20) (actual time=317.906..317.908 rows=99852.50 loops=2)
Buckets: 262144 (originally 8192) Batches: 1 (originally 1) Memory Usage: 15008kB
Buffers: shared hit=7353
-> Parallel Seq Scan on opt_orders o_1 (cost=0.00..13603.00 rows=2083 width=20) (actual time=0.033..241.593 rows=99852.50 loops=2)
Filter: (EXTRACT(year FROM order_date) = '2023'::numeric)
Rows Removed by Filter: 400148
Buffers: shared hit=7353
-> Hash (cost=36.00..36.00 rows=1000 width=14) (actual time=1.313..1.314 rows=1000.00 loops=2)
Buckets: 1024 Batches: 1 Memory Usage: 55kB
Buffers: shared hit=52
-> Seq Scan on opt_products p_1 (cost=0.00..36.00 rows=1000 width=14) (actual time=0.649..0.991 rows=1000.00 loops=2)
Buffers: shared hit=52
InitPlan 3
-> Aggregate (cost=17676.24..17676.25 rows=1 width=32) (actual time=591.381..599.418 rows=1.00 loops=1)
Buffers: shared hit=9319
-> Finalize GroupAggregate (cost=17664.09..17676.17 rows=5 width=18) (actual time=579.652..597.173 rows=5.00 loops=1)
Group Key: p_2.product_category
Buffers: shared hit=9319

-> Gather Merge (cost=17664.09..17676.08 rows=8 width=18) (actual time=576.520..597.126 rows=10.00 loops=1)
Workers Planned: 1
Workers Launched: 1
Buffers: shared hit=9319
-> Partial GroupAggregate (cost=16664.08..16675.17 rows=5 width=18) (actual time=503.528..518.212 rows=5.00 loops=2)
Group Key: p_2.product_category
Buffers: shared hit=9319
-> Sort (cost=16664.08..16667.76 rows=1473 width=10) (actual time=499.707..506.052 rows=49933.50 loops=2)
Sort Key: p_2.product_category
Sort Method: quicksort Memory: 1537kB
Buffers: shared hit=9319
Worker 0: Sort Method: quicksort Memory: 1537kB
-> Hash Join (cost=13677.54..16586.56 rows=1473 width=10) (actual time=370.674..459.437 rows=49933.50 loops=2)
Hash Cond: (o_2.product_id = p_2.product_id)
Buffers: shared hit=9311
-> Parallel Hash Join (cost=13629.04..16534.19 rows=1473 width=4) (actual time=368.564..436.721 rows=49933.50 loops=2)
Hash Cond: (c_2.id = o_2.client_id)
Buffers: shared hit=9259
-> Parallel Seq Scan on opt_clients c_2 (cost=0.00..2641.29 rows=29453 width=16) (actual time=0.054..23.950 rows=25012.00 loops=2)
Filter: ((status)::text = 'active'::text)
Rows Removed by Filter: 24988
Buffers: shared hit=1906

-> Parallel Hash (cost=13603.00..13603.00 rows=2083 width=20) (actual time=368.231..368.235 rows=99852.50 loops=2)
Buckets: 262144 (originally 8192) Batches: 1 (originally 1) Memory Usage: 15008kB
Buffers: shared hit=7353
-> Parallel Seq Scan on opt_orders o_2 (cost=0.00..13603.00 rows=2083 width=20) (actual time=0.038..275.380 rows=99852.50 loops=2)
Filter: (EXTRACT(year FROM order_date) = '2023'::numeric)
Rows Removed by Filter: 400148
Buffers: shared hit=7353
-> Hash (cost=36.00..36.00 rows=1000 width=14) (actual time=1.937..1.939 rows=1000.00 loops=2)
Buckets: 1024 Batches: 1 Memory Usage: 55kB
Buffers: shared hit=52
-> Seq Scan on opt_products p_2 (cost=0.00..36.00 rows=1000 width=14) (actual time=1.048..1.542 rows=1000.00 loops=2)
Buffers: shared hit=52
Planning:
Buffers: shared hit=33
Planning Time: 38.424 ms
Execution Time: 2685.754 ms